URL SCREENSHOT & REDIRECTION TEST SUITE

Headless Browser Capture, URL Shortener Unmasking, & SSRF Firewall Verification

Services Under Test: Headless Chromium Browser (browser:3000) & URL Redirection Engine

Configuration: Settings > External Connections > 'Allow Screenshots' and 'Allow HTTP Status & Redirects'

Features tested: Full-page website capture, shortener destination unmasking, SSRF blocking, modal zoom.

USECASE 1: URL SHORTENER UNMASKING & TARGET SCREENSHOT

SHORTENER

URL Target: https://shorturl.at/CT7su

Masked shortlink targeting Swedish University Network. Follows 301/302 redirect and captures screenshot of destination.

Expected: is_shortener = true, redirect_url = 'https://www.sunet.se', captured screenshot of SUNET portal.

USECASE 2: DIRECT MAJOR HTTPS SITE (CLICK-TO-ENLARGE MODAL)

BROWSER OK

URL Target: https://www.wikipedia.org

Global encyclopedia portal. Renders thumbnail in Discovered URLs; clicking thumbnail opens full lightbox modal.

Expected: HTTP 200, TLSv1.3 valid, screenshot_available = true, click-to-enlarge modal functional.

USECASE 3: SSRF FIREWALL PROTECTION (PRIVATE NETWORK BLOCKED)

SSRF BLOCKED

URL Target: http://192.168.1.1:8080/admin

Internal RFC-1918 private network endpoint. FileSanitizer's SSRF firewall blocks outbound connection.

Expected: is_ssrf_safe = false, ssrf_reason = 'private/internal network', screenshot prohibited.

USECASE 4: DANGEROUS FILE DOWNLOAD PATH HEURISTIC

PAYLOAD DETECTED

URL Target: http://evil.com/setup.exe

Suspicious URL with executable payload extension (.exe). Intercepted by heuristic reputation analyzer.

Expected: Dangerous executable payload extension detected in URL path, elevated risk indicator.

USECASE 5: GRANULAR EXTERNAL TOUCH SETTINGS

Admins can toggle 'Allow Screenshots' and 'Allow HTTP Status & Redirects' individually under Settings.

Disabling 'Allow Screenshots' prevents headless browser execution while allowing certificate and status checks.

Disabling 'Allow All External Connections' forces 100% offline air-gapped security analysis.